F78AA1 – December 2013 – Solutions

1. (a) $(\underline{I}a)^{(i)}_{\overline{n}|} = v + 2v^2 + \ldots + nv^n$ [1]

$\Rightarrow (1+i)(\underline{I}a)^{(i)}_{\overline{n}|} = 1 + 2v + \ldots + nv^{n-1}$ [1]

$\Rightarrow [(1+i)-1](\underline{I}a)^{(i)}_{\overline{n}|} = 1 + v + v^2 + \ldots + v^{n-1} - nv^n$ [1]

$\Rightarrow \quad i(\underline{I}a)^{(i)}_{\overline{n}|} = \ddot{a}^{(i)}_{\overline{n}|} - nv^n$

$\Rightarrow \quad (\underline{I}a)^{(i)}_{\overline{n}|} = \dfrac{\ddot{a}_{\overline{n}|} - nv^n}{i}$ [1]

[Total 4]

(b) (i) $1000\, s^{0.09}_{\overline{20}|} = 1000\, \dfrac{1.09^{20}-1}{0.09} = £51,160.12$ [2]

(ii) Define $j = 1.09^{1/2} - 1 = 0.044031$ (6 d.p.)

$1.09^{-5} \times 250 \times a^{0.044031}_{\overline{20}|} = £2,131.43$ [2]

(iii) Consider the p.v. of the payments made at integer times.

$225\, a^{(i)}_{\overline{10}|} + 25\, (\underline{I}a)^{(i)}_{\overline{10}|} = 2,213.7338.$ [2]

($a^{(i)}_{\overline{10}|} = 6.417658$, $(\underline{I}a)^{(i)}_{\overline{10}|} = 30.790431$)

Then the required p.v. is

$2,213.7338\left(1 + 1.09^{1/4} + 1.09^{1/2} + 1.09^{3/4}\right)$ [2]

$= £9,148.42.$ [Total 12]

2 (a) (i) $200{,}000 = X\left[a_{\overline{3}|}^{0.02} + 1.02^{-3}\, a_{\overline{27}|}^{0.05}\right]$ [1]

$2.883883 \quad 0.942322 \quad 14.643034$ [1]

$= £11{,}988.71$ [1]

(ii) Cap o/s = p.v. of future repayments

$= 11{,}988.71\; a_{\overline{20}|}^{0.05}$ [1]

$= £149{,}406.08.$ [1]

(b) After two years the loan has accumulated with interest at 8% to

$149{,}406.08\,(1.08)^2$ [1]

$= £174{,}267.25$ [1]

The new annual repayment will satisfy

$X' a_{\overline{18}|}^{0.05} = 174{,}267.25$ [1]

$\Rightarrow X' = £14{,}907.90$ [1]

[Total 9]

3 (a) – it must be real [1]
– it must be ≥ -1 [1]
– it must be the only real solution ≥ -1 [1]

(b) Steffensen's Theorem:

If the cash flows change sign exactly once the yield is guaranteed to exist. [2]

[Total 5]

4 (a) Define $j = 1.1^{1/2} - 1 = 0.048809$ [1]

$P = 4a_{\overline{40}|}^{0.048809} + 100(1.1)^{-20}$ [2]

$= 84.634875$ or £84.63 [1]

[Total 4]

(b) She gets net coupon of £3.20 per half year and [1]

redemption proceed of $100 - 0.3(100 - 84.63) = £95.39$

Hence the equation of value, using time unit half a year:

$84.63(1+i)^{40} - 3.2 s_{\overline{40}|}^{i} - 95.39 = 0$ [1]

Trial and error:

$f(0.04) = +6.839$

$f(0.03) = -60.608$ [2]

Linear interpolation:

$i^* \cong 0.03 + 0.01 \dfrac{60.608}{60.608 + 6.839}$

$= 0.039$ [1]

Hence the yield per annum is approximately ~~0.0795~~ 0.0795 or about 8%. [1].

[Total 6]

(c) If the net coupon is 6.4% per annum and the NRY is 10% [1]

p.a. the price must be > 100 so CGT will be payable

Hence:

$84.63(1.1)^{10} - 3.2 s_{\overline{20}|}^{0.048809} - [P - 0.3(P - 84.63)] = 0$ [2]

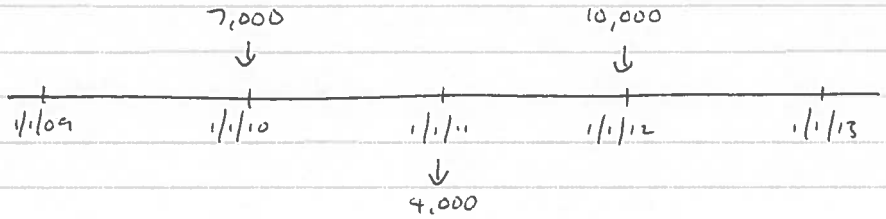
$\uparrow$ 219.51 $\quad\quad$ $\uparrow$ 104.49

$\Rightarrow 0.7P = 219.51 - 104.49 - 25.39 \Rightarrow P = £128.04$ [1].

[Total 4]

[Total 14]

5(a)



100,000	(105,000)	106,000	(108,000)	120,000	FUND

(a) Let i_1 be the yield in 2009-2010. Then

$$100(1+i_1)^2 + 7(1+i_1) - 106 = 0$$

Hence $(1+i_1)$ is a root of $100x^2 + 7x - 106 = 0$

$$\Rightarrow (1+i_1) = \frac{-7 \pm \sqrt{7^2 + 4(100)(106)}}{200}$$

$$\Rightarrow i_1 = -0.0048 \qquad [2]$$

Let i_2 be the yield in 2011-12. Then:

$$102(1+i_2)^2 + 10(1+i_2) - 120 = 0$$

Hence $(1+i_2)$ is a root of $102x^2 + 10x - 120 = 0$

$$\Rightarrow (1+i_2) = \frac{-10 \pm \sqrt{100 + 4(102)(120)}}{204}$$

$$\Rightarrow i_2 = +0.0367 \qquad [2]$$

Hence the LIRR is given by

$$(1+LIRR)^4 = 0.9952^2 (1.0367)^2 \qquad [1]$$

$$\Rightarrow LIRR = 0.01574. \qquad [1]$$

[Total 6].

(b) $$(1+TWRR)^4 = \frac{105}{100}\frac{106}{112}\frac{108}{102}\frac{120}{118} \qquad [1]$$

$$\Rightarrow TWRR = 0.01707 \qquad [1]$$

[Total 2]

[Total 8].

6. Calculate the profit under method A.

After one year the loan has accumulated to 10.7 m. [1]

It will be repaid at time $t+1$ where t is the smallest integer satisfying

$$10.7(1+i)^t \leq 4 s_{\overline{t}|}^{i} \qquad (i = 0.07) \qquad [1]$$

ie $$\frac{(1+i)^t - 1}{i(1+i)^t} \geq \frac{10.7}{4}$$

ie $$1 - v^t \geq 0.18725$$

ie $$v^t \leq 0.81275$$

ie $$(1+i)^t \geq 1.23039$$

ie $$t \geq \frac{\log(1.23039)}{\log(1.07)}$$

ie $$t \geq 3.064$$

Hence $$t = 4 \qquad [2]$$

When the loan is repaid the balance will be

$$4 s_{\overline{4}|}^{0.07} - 10.7(1.07)^4 = 3.734 \text{ m.} \qquad [1]$$

Hence the profit after six years is

$$3.734(1.05) + 4 - 2 = £5.92 \text{ m.} \qquad [1]$$

Calculate the profit under method B.

The investor must borrow £1m at $t=1$ to pay the [1]

first coupon.

Hence the profit at $t=2$ is

$$4 - 1 - 1.07 = 1.93 \text{ m.} \quad [1]$$

Hence the accumulated profit after six years is

$$1.93\underset{2.346}{(1.05)^4} + 3\,s\overline{\underset{12.930}{4}|}^{0.05} - 10 - 2 \quad [2]$$

$$= £3.276 \text{ m.} \quad [1]$$

Conclusion: the investor should choose method A. [1]

[Total 12]